EC3091D - Electronic Circuits Lab II

Experiment-8: Oscillator and Multivibrator Circuits

Aim:

1. To design, set up and study the operation of a Wien-bridge oscillator
2. To design, set up and study the operation of Multivibrator Circuits

Oscillator:

Circuit Diagram:

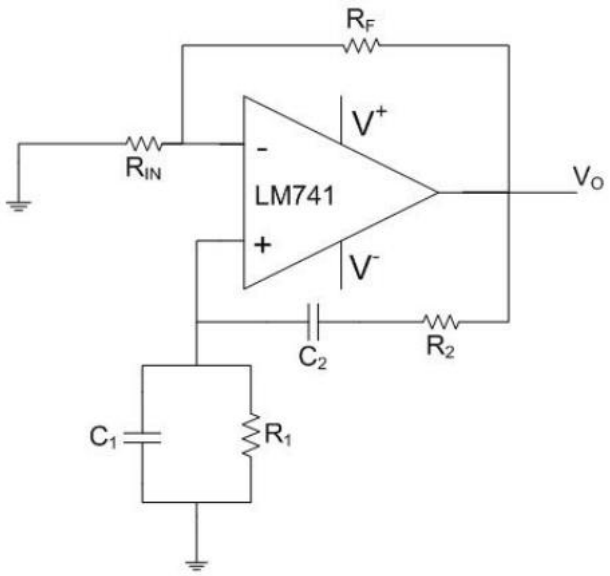


Figure 1

Procedure:

1. Set up the Wien-bridge oscillator as given in Figure 1.
2. Give suitable values for V^+ and V^-.
3. Vary the values of R_1, R_2, C_1 and C_2 and study the circuit.

Design:

1. Check the data sheet of the required Operational Amplifier (LM741) to find its electrical characteristics.
2. Select R_{IN}=1KΩ, R_F=1.5KΩ series with 1KΩ pot., R_1=R_2=10KΩ and C_1=C_2=0.1μF
3. Repeat by selecting R_1=R_2=1KΩ and C_1=C_2=0.22μF.

You may follow any design procedure, provided the circuit meets the specifications given.

Multivibrator Circuits:

Circuit Diagram:

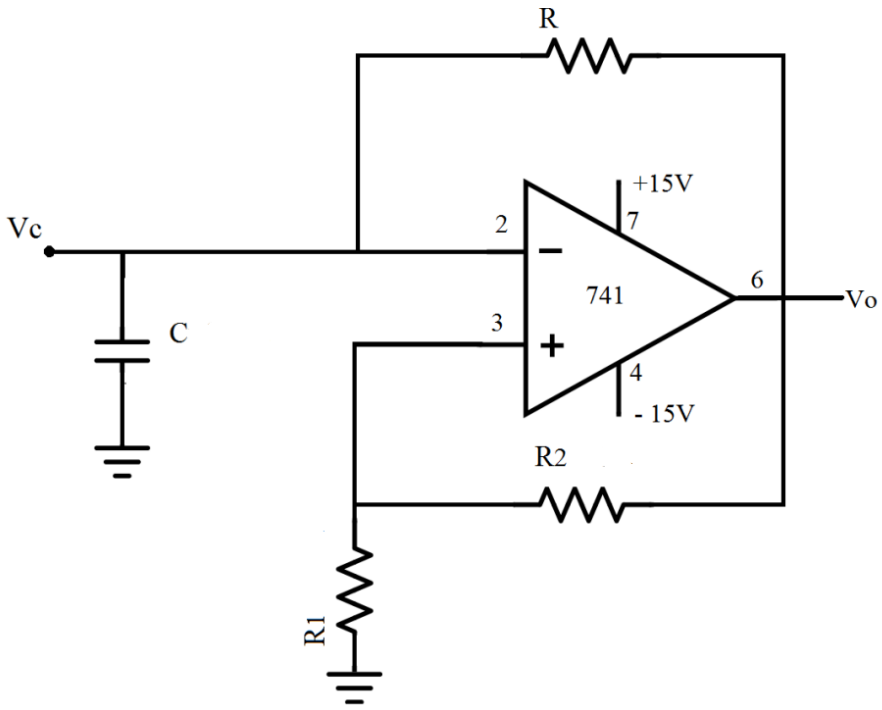


Figure 2 Astable Multivibrator

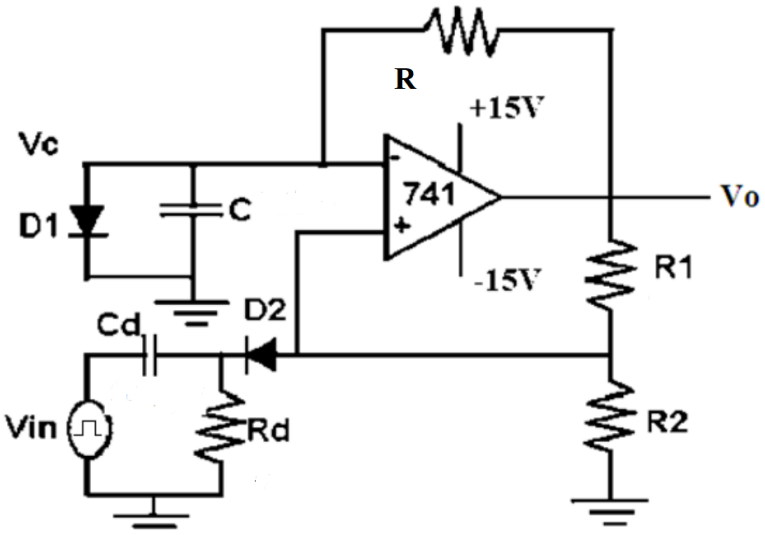


Figure 3 Monostable Multivibrator

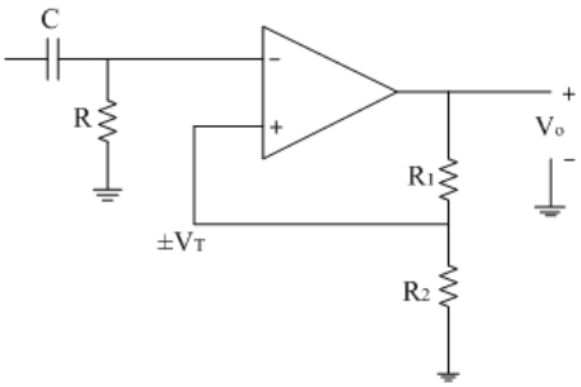


Figure 4 Bistable Multivibrator

Procedure:

Astable Multivibrator:

1. Configure the circuit as per the circuit diagram.
2. Set R_1 = 10kΩ, R_2 = 10kΩ, R = 4.7kΩ, and C = 0.1μF.
3. Calculate the frequency of the waveform.
4. Measure the obtained frequency of the waveform from the circuit.
5. Use a potentiometer instead of R and repeat steps 2 to 4 by varying the value of R.

Monostable Multivibrator:

1. Configure the circuit as per the circuit diagram.
2. Set R_1 = 10kΩ, R_2 = 10kΩ, R_d = 470Ω, C = 0.1μF, C_d = 0.01μF and R = 15kΩ.
3. Calculate the frequency of the waveform.
4. Measure the obtained frequency of the waveform from the circuit.
5. Use a potentiometer instead of R and repeat steps 2 to 4 by varying the value of R.

Bistable Multivibrator:

1. Configure the circuit as per the circuit diagram.
2. Set R_1 = R_2 = 10 kΩ, R = 100Ω and C = 0.1μF.
3. Show the transition from one state to another by applying a trigger pulse.

Reference:

1. Virtual Labs: An MoE Govt. of India Initiative, Analog Electronics Circuits

Online: http://vlabs.iitkgp.ac.in/psac/newlabs2020/vlabiitkgpAE/exp3/index.html#

2. S. Sedra, K.C. Smith, A.N. Chandorkar, Microelectronic Circuits, Oxford University Press, 2013